Gundla Suhas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.20
Gurukrupa Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	94.40%
Sri Vani High School	Class X (Board of Secondary Education, Telangana)	2021	10.00%

POSITIONS OF RESPONSIBILITY

Class Representative (CR), Dept. of Biotechnology and Bioinformatics

- Acted as the main link between the students and professors, helping manage class schedules, solving student issues, and sharing important department news to keep the class well-informed and connected.

SKILLS

Programming Languages: C, C++, Python, R, SQL

Frameworks & Libraries: TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Raylib

Bioinformatics Tools: BWA, SAMtools, GATK, FastQC, bcftools, IGV

Tools & Environments: Git, Github, VS Code, Linux

Core Concepts: Data Structures & Algorithms, Machine Learning, Deep Learning, Transfer Learning

RELEVANT COURSES

BT (Biotechnology & Bioinformatics): Big Data Biology and Biological Databases, R & Python for Biologists, Basic Bioinformatics, Biostatistics, Next Generation Sequencing, Biochemical Engineering.

CS (Computer Science): Data Structures & Applications, Intro to Programming (C), Artificial Intelligence.

MA (Mathematics): Probability & Random Variables, Linear Algebra, Calculus I, Calculus II.

LABORATORY EXPERIENCE

Basic Biotechnology Laboratory | *BTBM*

- Cultured and isolated *E. coli* (DH5) using **streak plating** and secondary inoculation, actively monitoring bacterial growth kinetics and plotting precise **growth curves** via OD spectrophotometry.
- Executed targeted **DNA PCR amplification**, **Agarose Gel Electrophoresis**, and **SDS-PAGE** analysis, accurately validating both plasmid and protein samples utilizing an advanced **Gel Documentation System**.
- Developed strong technical competencies in **aseptic techniques**, microbial culturing, molecular biology methodologies, and quantitative **experimental data interpretation** for robust biological analysis.

PROJECTS

Skin Cancer Classification using MobileNetV2

- Developed a deep learning pipeline for binary classification of **dermoscopic skin lesions**, leveraging **MobileNetV2 transfer learning** architectures applied to a comprehensive dataset of 3,297 clinical images.
- Designed robust **data preprocessing** workflows, implementing advanced **image augmentation** and strategic dataset splitting to significantly improve overarching model **generalization** capabilities and ensure high training stability.
- Evaluated model performance using comprehensive metrics, including detailed **Confusion Matrices** and Precision-Recall analysis, successfully achieving an impressive **83.94% test accuracy** alongside a **0.9056 ROC-AUC** score.
- Optimized model training by integrating essential callback mechanisms like **Early Stopping**, **Model Checkpointing**, and **ReduceLROnPlateau**, ensuring optimal weight retention and successfully validating on unseen dermoscopic data.

Next Generation Sequencing (NGS) Variant Calling Pipeline *(Course Project)*

- Developed an automated **variant calling pipeline** in **Linux** designed to thoroughly analyze and process paired normal and tumor sequencing datasets for a comprehensive academic course project.
- Assessed sequencing read quality using **FastQC**, indexed the reference genome, and executed efficient alignment of paired-end reads by leveraging the robust capabilities of the **BWA-MEM** algorithm.
- Processed genomic alignments through SAM-to-BAM format conversion, coordinate sorting, sequence indexing, and read group assignment utilizing the widely adopted **SAMtools** and **GATK** bioinformatics data processing frameworks.
- Identified somatic and germline variants using GATK **Mutect2** and **HaplotypeCaller**, comprehensively analyzed resulting VCF files with **bcftools**, and visually validated complex genomic alignments and mutations using **IGV**.

EXTRACURRICULAR

National Service Scheme (NSS), IIT Hyderabad *(2024-Present)*

- Active volunteer consistently participating in impactful **community service** initiatives, public awareness programs, educational workshops, and comprehensive **social outreach** events.

Chess

- Avid chess hobbyist who enjoys playing the game regularly, using it as a great way to improve **focus**, practice **thinking ahead**, and build basic **problem-solving** and **pattern-matching** skills.